

Weekly Progress Report

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Domain: Data Science and Machine Learning

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Week Ending: 01

Prediction of Agriculture Crop Production in India

1. Introduction

Agriculture is one of the most important sectors in India, supporting more than 50% of the population. Predicting crop production helps farmers and policymakers make better decisions.

2. Problem Statement

To build a machine learning model that predicts agricultural crop production based on factors like crop type, state, season, and cost of cultivation.

3. Dataset Description

The dataset contains information about agricultural production in India from 2001–2014.

Features:

- Crop
- Variety

- State
- Quantity
- Production (Target Variable)
- Season
- Unit
- Cost
- Recommended Zone

4. Data Preprocessing

- Handle missing values
- Encode categorical variables
- Remove duplicates
- Convert data types

5. Exploratory Data Analysis (EDA)

- Analyze production trends by state and crop
- Visualize relationships between cost and production

6. Model Building

We use multiple machine learning models:

- Linear Regression
- Random Forest Regressor

7. Python Implementation

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, r2_score
```

Load dataset

```
df = pd.read_csv('datafile.csv')
```

Handle missing values

```
df.dropna(inplace=True)
```

Encode categorical columns

```
le = LabelEncoder()
```

```
for col in ['Crop', 'Variety', 'State', 'Season', 'Recommended Z
one']:
```

```
    df[col] = le.fit_transform(df[col])
```

Features and target

```
X = df.drop('Production', axis=1)
```

```
y = df['Production']
```

Train-test split

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

Model

```
model = RandomForestRegressor()
```

```
model.fit(X_train, y_train)
```

Predictions

```
y_pred = model.predict(X_test)
```

```
# Evaluation
```

```
print('MAE:', mean_absolute_error(y_test, y_pred))
```

```
print('R2 Score:', r2_score(y_test, y_pred))
```

8. Results

- Random Forest performs well with high accuracy.
- Cost and quantity significantly impact production.

9. Conclusion

The model successfully predicts crop production. This can help farmers optimize crop choices and improve yield.

10. Future Scope

- Add weather data
- Use deep learning models
- Deploy as a web application using Streamlit

11. Mini Streamlit App

```
import streamlit as st
```

```
import numpy as np
```

```
st.title("Crop Production Prediction")
```

```
crop = st.number_input("Crop (encoded)")
```

```
state = st.number_input("State (encoded)")
```

```
cost = st.number_input("Cost")
```

```
if st.button("Predict"):
```

```
prediction = model.predict([[crop, state, cost]])  
st.write("Predicted Production:", prediction)
```

*“Include any additional comments or observations
regarding overall progress, collaboration, or notable
experiences during the week.”*